

Chapter 3

Intensity Transformation and Spatial Filtering

Part 2: Histogram Processing

3.3 Histogram Processing

- Image Size: MN
- r_k : k th intensity level
- n_k : the number of pixel in the Image with intensity r_k
- $p(r_k) = n_k / MN$: Normalized Histogram

3.3 Histogram Processing

$M=5, N=2, L-1=7$

1	2	3	2	0
2	3	3	3	2

r_k	0	1	2	3	4	5	6	7
n_k								
$p(r_k)$								

[Programming]

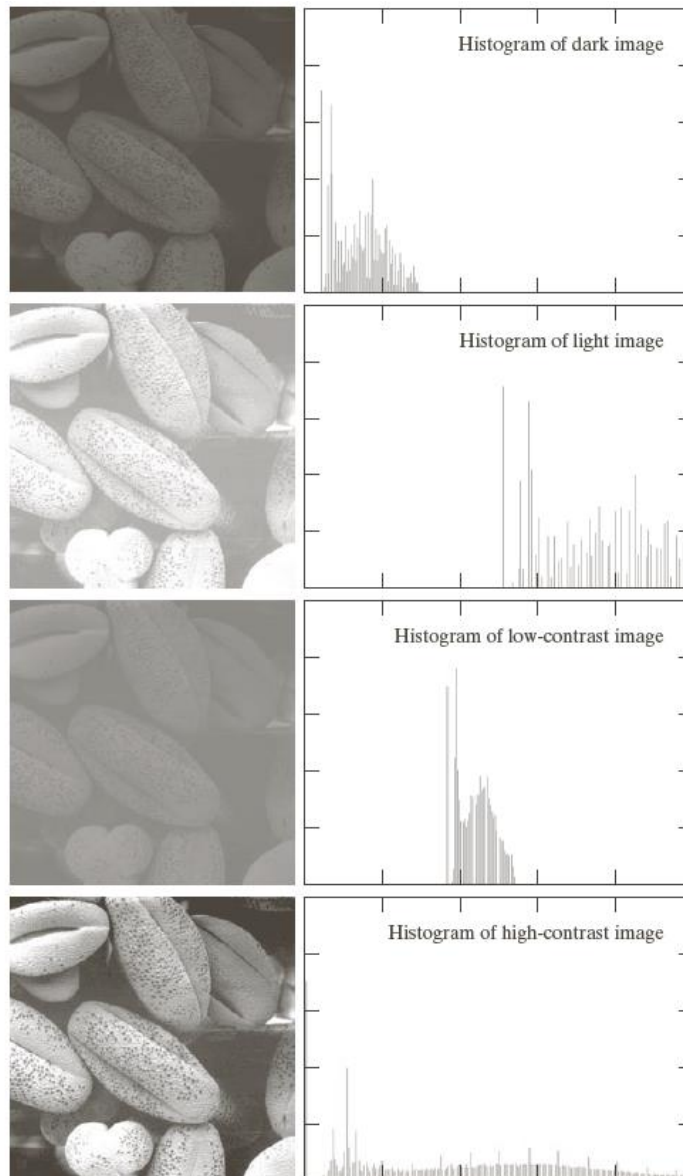


FIGURE 3.16 Four basic image types: dark, light, low contrast, high contrast, and their corresponding histograms.

3.3.1 Histogram Equalization

$$s = T(r) = (L - 1) \int_0^r p_r(w) dw$$

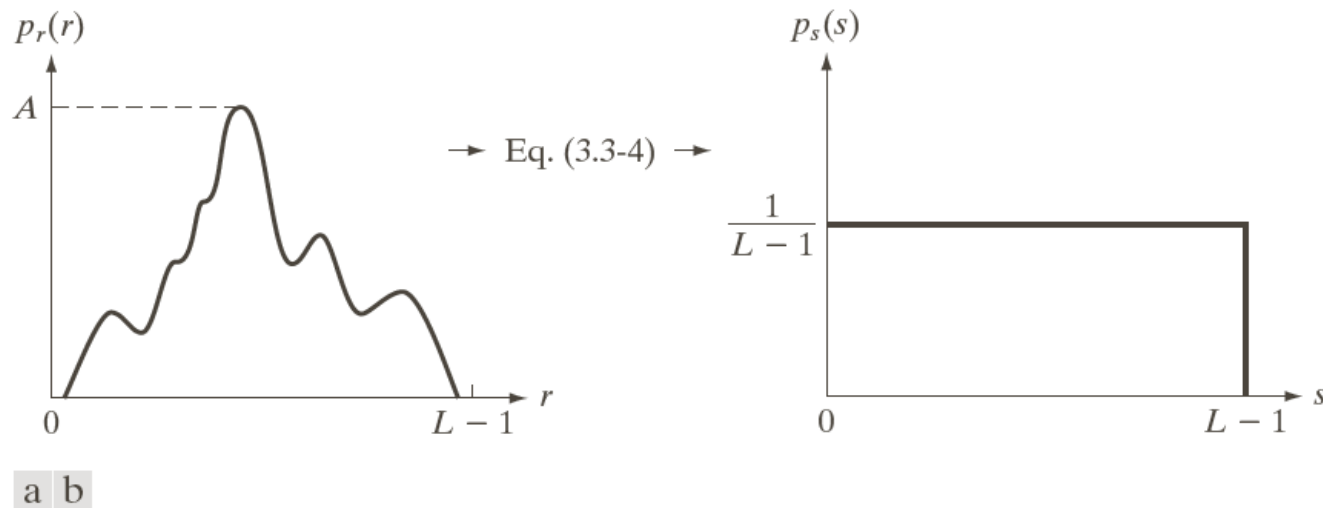


FIGURE 3.18 (a) An arbitrary PDF. (b) Result of applying the transformation in Eq. (3.3-4) to all intensity levels, r . The resulting intensities, s , have a uniform PDF, independently of the form of the PDF of the r 's.

3.3.1 Histogram Equalization

$$s = T(r) = (L - 1) \int_0^r p_r(w) dw$$

$$s = T(r_k) = (L - 1) \sum_{r_i=0}^{r_k} p(r_i)$$

3.3.1 Histogram Equalization

Input $M=5, N=2, L=7$

1	2	3	2	0
2	3	3	3	2

r_k	0	1	2	3	4	5	6	7
n_k	1	1	4	4	0	0	0	0
$p(r_k)$	0.1	0.1	0.4	0.4	0	0	0	0
$\sum P(r_k)$								
$(L-1) \sum P(r_k)$								

3.3.1 Histogram Equalization

- [Programming]